Specification Sheet
Dedicated Application for 3D Printing Requests

Managing requests, tracking activities, and improving workflow efficiency after expiration of the current Monday.com board

Field	Description
Project title	Dedicated application for managing 3D printing requests, tracking activities, and improving workflow efficiency.
Current situation	The current request process is managed through a Monday.com board that will expire or become unavailable.
Main risk	Loss of request visibility, historical traceability, ownership, status tracking, and workflow discipline.
Target result	A controlled application replacing Monday.com with request submission, validation, planning, execution tracking, dashboards, and historical archive.

1. Business Context

The current 3D printing request management process is handled through a Monday.com board. Its expiration creates a risk of losing operational visibility, request traceability, activity history, and follow-up discipline.

A dedicated application is required to replace the board and provide a more controlled, scalable, and production-oriented workflow for 3D printing request management.

2. Objective

Create a dedicated application that allows users to submit, validate, prioritize, execute, track, and close 3D printing requests with clear ownership, status visibility, deadlines, workload tracking, and performance indicators.

The application must reduce manual follow-up, avoid lost requests, support prioritization, and provide historical traceability of all 3D printing activities.

3. Scope

3.1 In Scope

- 3D printing request submission and request validation.
- Feasibility review, prioritization, planning, and assignment.
- Activity and status tracking with timestamps.
- File attachment management for CAD, STL, STEP, OBJ, 3MF, drawings, and sketches.
- Material, printer, printing duration, quality check, and post-processing tracking.
- Deadline management, notifications, dashboards, KPI monitoring, and historical archive.
- Migration or safe archiving of relevant Monday.com data.

4. Main Users and Responsibilities

User	Main needs / responsibilities
Requester	Submit a clear request, attach files/specifications, track status, provide missing information, and confirm completion.
3D Printing Coordinator / Process Owner	Validate request completeness, check feasibility, prioritize workload, assign owner, monitor backlog, escalate blocked requests, and review KPIs.

Technician / Operator	Execute assigned requests, access files and parameters, update progress, record material/printer/time/issues, and confirm completion.
Manager / Stakeholder	Review workload, service level, overdue requests, resource needs, bottlenecks, and performance trends.
Administrator	Manage users, permissions, statuses, priority rules, printers, materials, and master data.

5. Request Lifecycle

Standard workflow:

- Draft / New Request
- Submitted
- Completeness Check
- Feasibility Review
- Approved / Rejected / More Information Required
- Prioritized
- Planned
- Assigned
- In Progress
- Printed
- Post-Processing / Quality Check
- Ready for Pickup / Delivery
- Completed
- Archived

Alternative statuses: On Hold, Blocked, Cancelled, Rework Required, Waiting for Requester Input, Waiting for Material, Waiting for Machine Availability.

6. Required Request Fields

Category	Required fields
Request identification	Request ID, title, requester name, requester department, request date, required delivery date, project/customer reference if applicable.
Request description	Purpose, part description, quantity, functional/visual requirement, category, criticality, expected use environment.
Technical data	3D file, drawing/sketch, dimensions or scale, tolerance, surface finish, strength requirement, color, material preference, infill, layer height, orientation.
Planning and priority	Priority, reason for priority, requested due date, approved due date, estimated printing time, post-processing time, total lead time.
Execution data	Assigned technician, printer, material, batch/spool reference, start/end time, actual duration, post-processing, quality result, scrap/failed print, rework.
Closure data	Completion date, delivery confirmation, requester confirmation, final status, lessons learned, archive date.

8. Notifications and Alerts

- New submitted request.
- Missing information.
- Request approved or rejected.
- Assignment to technician.
- Due date approaching or overdue.
- Request blocked.
- Request completed.
- Request waiting for requester confirmation.

Notification channels may include email, in-app notification, and Microsoft Teams notification if applicable.

9. Dashboard Requirements

Dashboard	Indicators
Operational dashboard	Open requests, status split, priority split, requester/department split, assigned technician, overdue, blocked, printer workload, technician workload.
Performance dashboard	Average lead time, on-time completion rate, completed requests per week/month, rework rate, failed print rate, printer utilization, material consumption, backlog aging.
Management dashboard	Demand trend, capacity versus workload, top request categories, departments generating most requests, recurring issues, service level, resource bottlenecks.

10. Search and Filtering

Users should be able to search and filter by request ID, requester, department, status, priority, due date, assigned technician, printer, material, project reference, completion date, and blocked status.

11. Permissions and Access Control

Role	Permissions
Requester	Create requests, view own requests, edit before validation, respond to missing information, confirm receipt/completion.
Coordinator	View all requests, validate, approve, reject, prioritize, assign, reschedule, manage blocked requests, access dashboards.
Technician	View assigned requests, update execution status, record printing data, record issues/failures/completion.
Manager	View all requests and dashboards, export data, review workload and KPIs.
Administrator	Manage users, application settings, master data, exports, and archives.

12. Master Data to Configure

- Users and departments.
- Request categories, status list, and priority levels.
- Printers and materials.
- Standard lead-time rules.
- Rejection reasons and blocking reasons.
- Quality check criteria.

14. Reporting and Export

The application should allow Excel or CSV export for open requests, completed requests, overdue requests, workload by technician, workload by printer, KPI history, and full historical archive.

15. Efficiency Improvements Expected

- Reduce manual follow-up.
- Avoid lost or forgotten requests.
- Clarify request completeness before execution.
- Improve prioritization and visibility of workload.
- Reduce delays caused by missing information.
- Improve traceability of decisions and status changes.
- Provide performance data for resource planning.

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19. Acceptance Criteria

- Users can submit complete 3D printing requests with attachments.
- Coordinator can validate, approve, reject, prioritize, and assign requests.
- Technicians can update execution progress and completion data.
- Managers can see open, overdue, blocked, and completed requests.
- All requests have a clear status, owner, and due date.
- Historical data from Monday.com is either migrated or safely archived.
- KPIs are available for lead time, on-time completion, backlog, and rework/failure rate.
- Users can export request data.
- The system prevents closure without completion confirmation.
- The workflow is simple enough to replace Monday.com without operational disruption.

20. Proposed Implementation Phases

Phase	Main activities
1 — Current state capture	Review current Monday.com board, columns, statuses, automations, attachments, users, and pain points.
2 — MVP design	Confirm workflow, fields, roles, permissions, notifications, and dashboard requirements.
3 — Build and test	Build MVP, test with historical requests, validate user roles, notifications, exports, and dashboards.
4 — Migration and go-live	Export Monday.com data, import or archive history, train users, launch application, monitor first usage.
5 — Stabilization and improvement	Review adoption, close gaps, improve reporting, and add advanced features only after workflow is stable.

21. Open Points to Clarify

- Who is the application owner?
- Who are the main requesters?
- How many requests are handled per week or month?
- What data currently exists in Monday.com?
- Are attachments critical to migrate?
- Is the application internal only or accessible by external users?
- Should the application integrate with Microsoft Teams, SharePoint, ERP, or printer software?
- Is cost tracking required from day one?
- Is material stock tracking required from day one?
- What is the exact expiration date of the current Monday.com board?